

Joshua Panicker

[GitHub](#) | [LinkedIn](#) | [Website](#) | US Citizen

EDUCATION

California Polytechnic State University, San Luis Obispo, CA
Bachelor of Science, Computer Science

Expected: June 2029
GPA: 3.55/4.0

Relevant Coursework: Data Structures, Design & Analysis of Algorithms, Object-Oriented Programming & Design, Systems Programming, Computer Organization

In Progress: Programming Languages, Computer Security, Database Management Systems

Additional Coursework:

De Anza College: Operating Systems, C++ Programming, Java Programming & Problem Solving, Python Programming

Foothill College: Discrete Mathematics

TECHNICAL SKILLS

Languages: Python, Java, C, C++, TypeScript, JavaScript, SQL, RISC-V Assembly

Web Development: React, Next.js, Tailwind CSS

Databases: PostgreSQL, SQLite, MongoDB

Tools & Platforms: Git, GitHub, Supabase, Vercel, Vite, Unix, Cursor

AI: Retrieval-Augmented Generation (RAG), LLM Integration, Anthropic Claude API

EXPERIENCE

CodeBox - Director of Engineering, Projects / Co-Lead

(Cal Poly SLO computer science club)

Oct 2025 - Present

- Oversee **6–8 distinct project teams**, guiding each team's technical lead on architecture, implementation priorities, and project delivery.
- Led architecture and full-stack development for an academic resource-sharing platform with an **8-person engineering team**, defining API patterns, database schemas, and implementation priorities.
- Enforced code quality and architectural consistency through full-stack pull request reviews, Git commit conventions, and technical documentation.
- Designed a software development bootcamp curriculum for prospective club members.
- Organize hackathons for the Cal Poly student developer community to support collaborative software development.

Hack4Impact Software Developer

(Student-run organization developing projects for local nonprofits)

Oct 2025 - May 2026

- **Replaced manual community event publishing workflows** for Spokes by developing a full-stack event discovery and management platform.
- Enabled event creation, management, filtering, and saving by implementing RESTful API endpoints in Next.js and TypeScript.
- Implemented event data persistence and validation by designing MongoDB schemas and integrating Mongoose.

PROJECTS

Stockbrook - [AI Stock Screening & Portfolio Dashboard](#)

April 2026 - Present

- **Screened 5K+ equities** from a universe of 5.7K+ tracked symbols by building a configurable engine that evaluates P/E ratios, revenue growth, and 52-week positioning against JSON-defined rules.
 - Grounded AI-generated verdicts in SEC filings by building a RAG pipeline indexing **3.8K filing chunks across 75 tickers** using SQLite full-text search and source attribution.
 - Built a FastAPI backend with **46 REST endpoints** and delivered Claude responses token by token to the React frontend using server-sent events.
 - Enabled users to customize financial dashboards through a responsive React and TypeScript interface with drag-and-drop layouts and TradingView/Recharts visualizations.
- Tech:** Python, FastAPI, Anthropic Claude API, Yahoo Finance API (yfinance), React, TypeScript, Vite, Tailwind CSS

Polypages - [Campus Note-Sharing Platform](#)

Nov 2025 - May 2026

- **Served 50+ users** by building and deploying a campus-exclusive note-sharing platform with calpoly.edu email verification, Next.js, Supabase, and Vercel.
 - Implemented a credit-based exchange for academic resources using PostgreSQL triggers and RPCs, allowing users to earn credits through approved uploads and upvotes and spend them to unlock PDFs.
 - Restricted PDF access through server-side authorization and Supabase Storage signed URLs with 1-hour expiration.
 - Enabled course-based resource discovery using department filters, PostgreSQL trigram indexes for fuzzy search, and personalized enrolled-course prioritization.
- Tech:** Next.js 16, TypeScript, React 19, Supabase, PostgreSQL, Tailwind CSS, Vercel